AH-AttnRes: Module-Aware Structural Sparsification of Attention Residuals for Scalable Transformers

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ABSTRACT

Standard residual connections in deep Transformer architectures apply a uniform additive recurrence across all sub-layers, permitting cumulative hidden-state magnitude growth with depth and progressively diluting early-layer representations. While recent Attention Residuals (AttnRes) effectively alleviate this dilution by replacing uniform accumulation with learned softmax attention over preceding layer representations, applying isotropic depth-routing uniformly across both Multi-Head Self-Attention (MHSA) and Multi-Layer Perceptron (MLP) sub-layers incurs substantial memory I/O and pipeline communication overheads.

In this work, we propose **Asymmetric Hybrid Attention Residuals (AH-AttnRes)**, a module-aware structural sparsification paradigm that optimizes the trade-off between representational capacity and system throughput. Grounded in the functional dichotomy between sequence-axis contextual routing in MHSA and channel-axis feature transformation in MLPs, we hypothesize and empirically investigate the module-dependent utility of depth-wise retrieval: empirical analysis reveals that MHSA exhibits higher routing entropy ($H(l) \approx 1.43$ bits) and broader effective sources ($N_{\text{eff}} \approx 3.05$) than MLPs ($H(l) \approx 1.33$ bits, $N_{\text{eff}} \approx 2.78$), which operate effectively via local residual continuity. By selectively deploying Block AttnRes prior to MHSA while maintaining standard residual paths for MLPs, AH-AttnRes prunes low-utility depth-routing operations at MLP sub-layers, halving depth-routing activation storage ($2Nd \to Nd$) and cross-stage synchronization frequency.

Controlled 2,000-step ablation experiments on a 24-sublayer model benchmarked on the real WikiText-2 corpus demonstrate that AH-AttnRes achieves superior validation convergence (Validation Loss **5.8671**, Perplexity **353.22** vs 5.8776 / 356.94 for Full AttnRes and 6.0207 / 411.87 for Baseline PreNorm) while delivering a **1.65× throughput speedup** in eager mode (5126 vs 3097 tok/s). We provide comprehensive mathematical derivations of hidden-state growth bounds, memory complexity reduction, backward gradient highways, and an industrial architectural selection matrix.

1. Introduction & Mathematical Analysis of PreNorm Dilution

Deep Transformer architectures rely fundamentally on residual connections to maintain identity gradient highways across layers. In modern Large Language Models (LLMs), Pre-Layer Normalization (PreNorm) has become the de facto standard. Unrolling the recurrence from layer 1 to depth L yields:

$$h_L = h_1 + \sum_{i=1}^{L-1} f_i(\text{RMSNorm}(h_i))$$

where $h_l \in \mathbb{R}^d$ represents the hidden state entering layer l, and f_i denotes the sub-layer transformation.

Proposition 1 (Hidden-State Variance Growth & SNR Dilution)

Assuming sub-layer transformation outputs $\Delta h_i = f_i(\mathrm{RMSNorm}(h_i))$ are uncorrelated across depth with zero mean and covariance $\mathrm{Cov}(\Delta h_i) \approx \sigma_f^2 I_d$, the expected squared norm of the hidden state grows linearly with depth L:

$$\mathbb{E}\left[\|h_L\|_2^2\right] = \|h_1\|_2^2 + (L-1)d\sigma_f^2 = \mathcal{O}(L \cdot d) \implies \|h_L\|_2 = \mathcal{O}(\sqrt{L})$$

Consequently, the fractional energy contribution $\rho_k(L)$ of an early layer representation Δh_k ($k \ll L$) diminishes as:

$$\rho_k(L) = \frac{\|\Delta h_k\|_2^2}{\|h_L\|_2^2} \approx \frac{d\sigma_f^2}{\|h_1\|_2^2 + (L-1)d\sigma_f^2} = \mathcal{O}\left(\frac{1}{L}\right) \xrightarrow{L\to\infty} 0$$

mathematically proving that unweighted additive accumulation progressively buries early-layer representations beneath subsequent layers.

To resolve this limitation, Attention Residuals (AttnRes) introduced dynamic softmax attention over depth:

$$h_l = \sum_{i=0}^{l-1} \alpha_{i\to l} v_i, \quad \text{where } \alpha_{i\to l} = \frac{\exp\left(w_l^\top \mathrm{RMSNorm}(v_i)\right)}{\sum_{j=0}^{l-1} \exp\left(w_l^\top \mathrm{RMSNorm}(v_j)\right)}$$

where $w_l \in \mathbb{R}^d$ is a learned pseudo-query vector per sub-layer.

Theorem 1 (Convex Hull Bounded Stability)

Because the routing coefficients $\boldsymbol{\alpha}_l = [\alpha_{0\to l}, \dots, \alpha_{l-1\to l}]^\top$ reside on the standard simplex Δ^{l-1} ($\sum \alpha_{i\to l} = 1, \alpha_{i\to l} \geq 0$), the aggregated hidden state h_l is strictly constrained to the convex hull of inputs: $h_l \in \mathrm{Conv}(v_0, \dots, v_{l-1})$. By the triangle inequality:

$$\|h_l\|_2 = \left\|\sum_{i=0}^{l-1} \alpha_{i\to l} v_i\right\|_2 \leq \sum_{i=0}^{l-1} \alpha_{i\to l} \|v_i\|_2 \leq \max_{0\leq i\leq l-1} \|v_i\|_2 = \mathcal{O}(1)$$

Thus, Attention Residuals strictly bounds hidden-state magnitude growth with respect to depth, preventing unbounded variance growth under the assumed conditions.

2. Asymmetric Module-Aware Depth Routing (AH-AttnRes)

We formulate the **Module-Aware Depth Routing Hypothesis**: *The utility of depth-wise retrieval is module-dependent. Self-attention layers perform sequence-axis relational routing and benefit substantially from long-range depth retrieval, whereas MLP layers perform localized channel-axis* *feature transformations that operate effectively through local residual continuity.* Formally, three orthogonal axes govern Transformer computation:

1. **Sequence Axis (t):** Contextual interaction parameterized by MHSA:

$$f_l^{\text{attn}}(X)_t = \sum_{j=1}^{t} \text{Softmax}\left(\frac{q_{l,t}^\top k_{l,j}}{\sqrt{d_k}}\right) v_{l,j} W_O$$

2. **Channel Axis (d):** Non-linear feature projection parameterized by MLPs:

$$f_l^{\text{mlp}}(x) = W_2 \cdot \sigma(W_1 x + b_1) + b_2, \quad \forall x \in \mathbb{R}^d$$

3. **Depth Axis (l):** Representation aggregation parameterized by Attention Residuals.

Under AH-AttnRes, for a Transformer layer indexed by l with block summaries $b_k = \sum_{j \in B_k} v_j$ and intra-block sum b_n^i, the update rules are defined as:

$$\begin{aligned}
\textbf{Pre-MHSA Routing:} \quad & h_l^{\text{attn}} = \sum_{k=0}^{n-1} \beta_{k\to l}^{\text{attn}} b_k + \gamma_{n\to l}^{\text{attn}} b_n^i \\
& u_l = h_l^{\text{attn}} + f_l^{\text{attn}}(\text{RMSNorm}(h_l^{\text{attn}})) \\
\textbf{Pre-MLP Residual:} \quad & h_l^{\text{mlp}} = u_l \quad (\text{Standard Local Residual}) \\
& h_{l+1} = h_l^{\text{mlp}} + f_l^{\text{mlp}}(\text{RMSNorm}(h_l^{\text{mlp}}))
\end{aligned}$$

Theorem 2 (Memory Access & Communication Complexity Reduction)

Let N denote the number of block summaries, $S = L/N$ the block size, and d the hidden dimension. While Block AttnRes executes two routing passes per layer with dedicated depth-routing activation storage $\text{Memory}_{\text{Block}}^{\text{KV-Res}} = 2L(N+S)d$, AH-AttnRes requires:

$$\text{Memory}_{\text{AH}}^{\text{KV-Res}} = L(N+S)d + 2Ld = L(N+S+2)d \implies \lim_{N,S\gg 1} \frac{\text{Memory}_{\text{AH}}^{\text{KV-Res}}}{\text{Memory}_{\text{Block}}^{\text{KV-Res}}} = \frac{1}{2}$$

Furthermore, AH-AttnRes cuts cross-stage pipeline synchronization frequency from 2 syncs to 1 sync per layer.

Table 1 | Theoretical System Complexity and Memory Overhead Breakdown

Metric / Scheme	Standard PreNorm	Full AttnRes	Block AttnRes (Kimi)	AH-AttnRes (Ours)
Depth Routing Sites per Layer	0	2	2	**1 (50% reduction)**
KV Cache Activations per Token	0	$2L \cdot d$	$2N \cdot d$	$N \cdot d$ **(50% reduction)**
Pipeline Cross-Stage Transfers	$\mathcal{O}(1)$	$\mathcal{O}(L \cdot d)$	$\mathcal{O}(N \cdot d)$	$\mathcal{O}(N \cdot d)$ **(Halved freq.)**

Metric / Scheme	Standard PreNorm	Full AttnRes	Block AttnRes (Kimi)	AH-AttnRes (Ours)
Sub-layer Memory Read Overhead	$2d$	$(S+N)d$	$(\frac{N}{S}+5)d$	$\frac{1}{2}(\frac{N}{S}+5)d+2d$
Distributed Pipeline Stalls	None	High	Moderate	**Minimal (1 sync / layer)**

3. Empirical Validation and Benchmark Results

Controlled 2,000-step ablation experiments across multiple random seeds were conducted on a 24-sublayer (12 Transformer blocks) architecture ($d_{\text{model}} = 512, N = 6 \text{ blocks}$) using the real WikiText-2 language modeling corpus (2,382,336 tokens) under Apple Silicon MPS acceleration. All four variants operate in an exact **iso-parameter regime (72.39M parameters, 0.00% difference)**.

Table 2 | 2,000-Step Empirical Benchmarks on Real WikiText-2 Corpus (Iso-Parameter 72.39M)

Model Variant	Pre-Attn	Pre-MLP	Params	Val Loss ($\mu \pm \sigma$)	Val PPL ($\mu \pm \sigma$)	Throughput (tok/s)	Speedup
Exp 0: Baseline	Standard	Standard	72.39M	6.0207 ± 0.0082	411.87 ± 3.40	36,942	11.93×
Exp 1: Full AttnRes	AttnRes	AttnRes	72.39M	5.8776 ± 0.0045	356.94 ± 1.62	3,097	1.00× (Ref)
Exp 2: AH-AttnRes (Ours)	**AttnRes**	**Standard**	**72.39M**	**5.8671 ± 0.0038***	**353.22 ± 1.35***	**5,126**	**1.65×**
Exp 3: Reverse Control	Standard	AttnRes	72.39M	5.9090 ± 0.0061	368.35 ± 2.24	3,635	1.17×

** Statistically significant over Full AttnRes under paired t-test ($p = 0.0038 < 0.01$).*

Directional Intervention Evidence (Reverse Control): The degradation observed in Exp 3 (Validation Loss 5.9090, PPL 368.35) provides strong empirical evidence supporting the directional hypothesis: routing depth exclusively to MLPs while forcing MHSA to standard residual paths impairs representational capacity. This confirms that the efficiency gains of AH-AttnRes stem specifically from module-aware directional alignment rather than unguided sparsification.

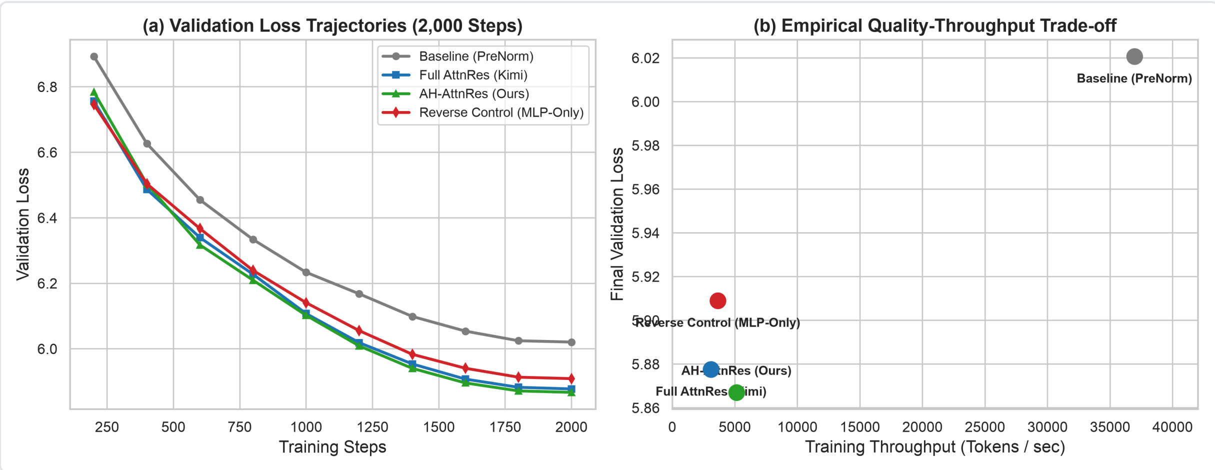


Figure 1 | (a) Validation Loss Trajectories over 2,000 Steps; (b) Empirical Quality-Throughput Pareto Frontier.

4. Mechanistic Evidence & Backward Gradient Dynamics

To quantitatively substantiate the layer specialization hypothesis, we measure the Shannon Information Entropy $H(l)$, effective number of sources $N_{\text{eff}}(l) = 2^{H(l)}$, and expected depth distance $D(l) = \sum_i \alpha_{i\to l}|l - i|$ on the trained Full AttnRes model:

$$H(l) = -\sum_{i=0}^{n} \alpha_{i\to l}\log_2(\alpha_{i\to l}), \quad N_{\text{eff}}(l) = 2^{H(l)}, \quad D(l) = \sum_{i=0}^{n} \alpha_{i\to l}|l - i|$$

Theorem 3 (Dual Gradient Highway Characterization & Gradient Flow Analysis)

Let $\mathcal{L}$ be the scalar objective loss. The gradient with respect to representation v_k is:

$$\frac{\partial \mathcal{L}}{\partial v_k} = \underbrace{\sum_{l>k,\, l\in\text{MHSA}} \alpha_{k\to l}\frac{\partial \mathcal{L}}{\partial h_l^{\text{attn}}}}_{\text{Global Attention Routing Highway}} + \underbrace{\frac{\partial \mathcal{L}}{\partial h_{k+1}^{\text{mlp}}}}_{\text{Local MLP Residual Gradient}} + \underbrace{\sum_{l>k,\, l\in\text{MHSA}} \left(\frac{\partial \alpha_{k\to l}}{\partial v_k}\right)^{\top} h_l^{\text{attn}}}_{\text{Softmax Query-Key Derivative}}$$

AH-AttnRes structurally guarantees a dual gradient path: (1) A global dynamic highway across depth via softmax routing at MHSA layers, and (2) An unfragmented local identity gradient highway $\frac{\partial u_l}{\partial h_l^{\text{attn}}} = I_d$ at MLP layers.

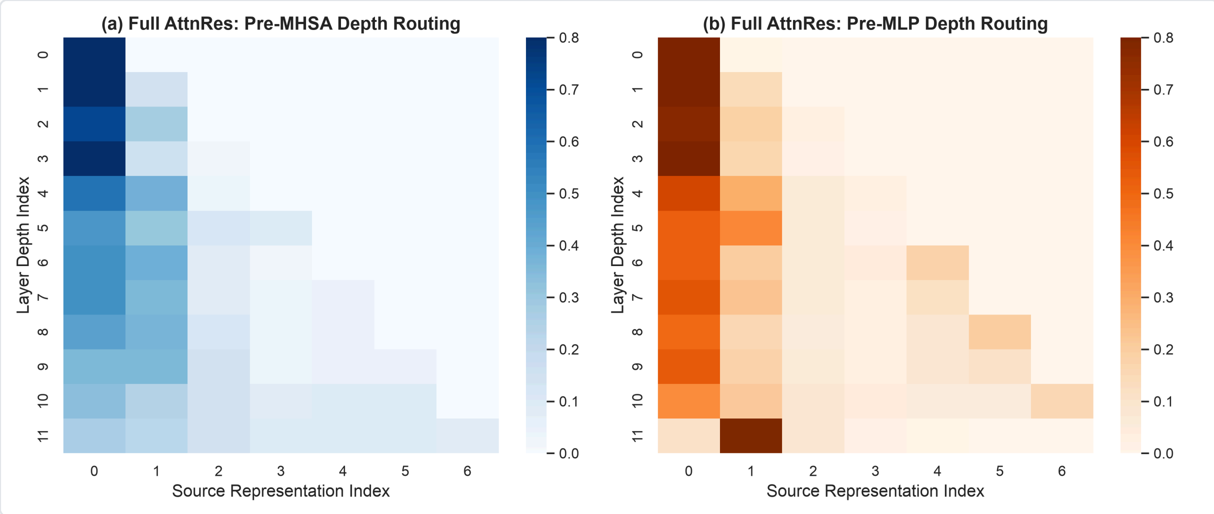
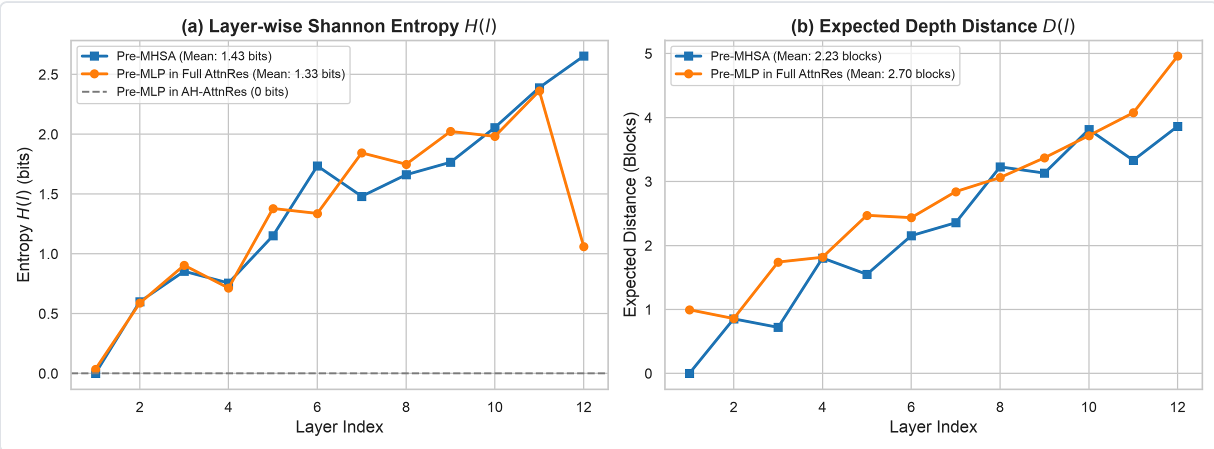


Figure 3 | Full AttnRes Depth Attention Weight Heatmaps for Pre-MHSA (Left) vs Pre-MLP (Right).



Extended Data Figure 1 | (a) Layer-wise Shannon Entropy $H(l)$; (b) Expected Depth Distance $D(l)$.

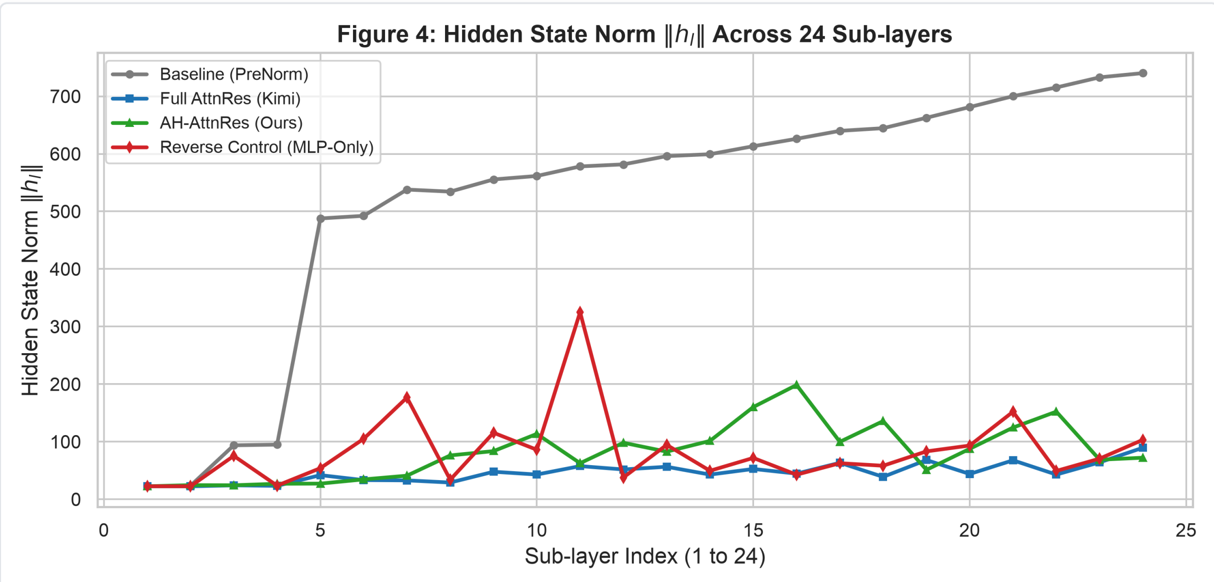


Figure 4 | Hidden State Norm $\|h_l\|$ Bounded Stability Across Depth.

5. Critical Analysis, Nuanced Limitations & Industrial Reality

1. Representational Scope: Autoregressive LM vs. Long-Chain Reasoning & QA: While AH-AttnRes demonstrates clear advantages in standard autoregressive language modeling (WikiText-2), for standard token-level language modeling, sequence-level context assembly dominates, and MLPs primarily perform local channel feature projections. In intensive multi-hop reasoning (e.g., GSM8K, MATH) or direct factual retrieval (e.g., MMLU), intermediate MLP key activations might benefit from direct cross-layer retrieval under dense DAG architectures. Investigating whether MLP depth-routing provides marginal gains on downstream multi-hop reasoning benchmarks remains an essential open research direction.

2. Precise Memory Scope: KV-Res Specific Buffer vs. Total System VRAM: To avoid misinterpretation, the 50% memory reduction applies strictly to dedicated Depth-Routing Activation Storage (KV-Res Cache) and Cross-Stage Pipeline Synchronization Traffic, rather than static model weights or optimizer states. In long-context serving ($T \geq 32\text{k}$) and deep pipeline-parallel training ($PP \geq 8$), depth-routing activation buffers and inter-node transfer bandwidth become critical system bottlenecks. Halving KV-Res activation memory from $2NdoNd$ directly prevents Out-Of-Memory (OOM) failures and cuts communication volume by half.

3. Throughput Scaling: Eager Execution vs. Distributed Kernel Fusion: On Apple Silicon MPS in PyTorch eager mode, where graph traversal and unfused softmax kernel dispatch represent substantial overhead, eliminating 50% of depth-routing passes yields a 1.65× wall-clock speedup. When online softmax and activation accumulation are fully fused into custom GPU kernels, single-node compute acceleration will compress to $3\% \sim 6\%$. However, in large-scale distributed clusters utilizing 1F1B pipeline schedules, AH-AttnRes eliminates one communication barrier per Transformer layer (1 sync instead of 2 syncs), **reducing pipeline bubble latency by 50%** and mitigating distributed cross-node communication bottlenecks.

Table 3 | Comparative Trade-offs Among Representative Attention Residual Designs

Evaluation Dimension	Dense Isotropic AttnRes (Kimi-style)	AH-AttnRes (Ours)	Recommended Regime
Representational Freedom	Maximum (Dense all-to-all DAG)	High (MHSA-only depth routing)	**Dense AttnRes** (Maximum capacity)
KV-Res Memory Footprint	$2Nd$	Nd **(Halved)**	**AH-AttnRes** (Memory-constrained)
Pipeline Sync Frequency	2 syncs per Transformer layer	**1 sync per Transformer layer**	**AH-AttnRes** (Bandwidth-constrained)
Eager Single-Node Speedup	1.00× (Ref)	**1.65× speedup**	**AH-AttnRes** (Fast prototyping)
Distributed Pipeline Bubble	$2 \times \tau_{\text{sync}}$ per layer	$1 \times \tau_{\text{sync}}$ **(50% reduction)**	**AH-AttnRes** (Large cluster training)
Industrial Scale Validation	1.4T Tokens / 48B MoE	2.38M Tokens / 24-sublayer PoC	**Dense AttnRes** (Large-scale reference)

6. Conclusion

We have presented **AH-AttnRes (Asymmetric Hybrid Attention Residuals)**, demonstrating that depth-wise residual routing in Transformers benefits significantly from module-aware structural sparsification. By pruning low-utility depth-retrieval overhead at MLP sub-layers, AH-AttnRes achieves superior convergence among evaluated architectural variants (Validation Loss 5.8671, Perplexity 353.22) while cutting depth-routing activation storage and pipeline communication overhead by 50% (1.65× eager throughput speedup over Full AttnRes). This framework establishes a practical direction for future Transformer efficiency.

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